

capacity of the CD, the manufacturer of the medium, supported writing speeds, last possible address where data can be written, etc. Only a CD-Writer can read the pre-groove, since this information is only needed when writing a CD. When an application starts, it checks whether the ATIP information is present. If it is, then the program assumes that it is running on a CD-Writer and will refuse to run. If however, the same CD is put in a normal CD-ROM drive, the ATIP information will be invisible to the application and it will run normally. A disc using this means is thus prevented from running on CD-Writers.

Save thy DVD

DVD content has been relatively safe till date. A copied DVD movie runs in excess of 1 GB of space. Compared to the 3 to 5 MB that a typical audio file takes up when compressed as MP3, it is an unruly size to say the least. Thus, movie distri-

and can thus continue to tag a protected content.

Content Scrambling System (CSS):

This is a data encryption and authentication scheme intended to prevent copying video files directly from DVD-Video discs. The content is stored in a scrambled state on a DVD-ROM disc. Every DVD player is coded with a small set of player keys which identify that player as an authorised CSS playback mechanism. These keys are provided by the DVD licensing authorities for a sizeable fee. Once a DVD disc is inserted into a DVD player, the disc and the player exchange information and if the player key is valid, the disc unlocks a title key. The title key is then used to unscramble the data on the DVD. The actual process of content protection thus involves the validation of a DVD player, rather than that of the medium.

info BBC reports that Napster, the online music swapping service, had an estimated 60 million users at the height of its use

RealOne. Nor could the digital files be played on anything other than a Windows operating system. Other copy protection efforts have run into trouble with the public.

Natalie Imbruglia's music album, *White Lilies Island*, used an encoding process to protect its content. This process corrupted the music stored on the disc. A hi-fi CD player's error correction mechanism was meant to compensate for the corrupt data and recreate the sound to a level that was claimed to be undetectable by the listener. Put the CD into a PC, however, and the drive would pick up the corrupt content and claim the disc to be unreadable. The furore created by this move forced BMG, the distributors of the album, to issue unhappy consumers with replacement discs that lacked the offending technology.

Cases like these are not limited to the music industry. Game distributors

“The consumers started getting back saying that it doesn't play on car CD players and several types of normal players. We were really shocked...”



— Mathias Immel,
Head of product coordination and new media at BMG, Germany

bution over the peer-to-peer networks that lace the Internet is not as brisk an activity as music-swapping. The movie industry thus has some breathing space to come up with a viable solution before things take a turn for the worse for them.

Watermarking: DVD audio is copy-right protected by a digital watermark. Signal processing technology is used to apply a digital signature (and optional encryption keys) to the audio. Watermarking permanently marks each digital audio frame with noise that is supposedly undetectable by human ears. An attempt to copy a watermarked DVD will fail in recreating the signature. Equipment that support this technique will recognise copied audio (by virtue of the absence of added noise) and refuse to play it. Watermark signatures can be recognised by playback and recording equipment to prevent copying. Even when the signal is transmitted via digital or analog connections or is subjected to video processing, the watermark is retained

The fallout

What can be made can just as easily be broken. There is a silent war going on between the content creators and the crackers. While the former strive to keep one step ahead of the race, the latter would like nothing better than to rub the noses of the giants in the dirt, by defeating their protection schemes. Caught between the two is the end-user. These efforts for media control have had an immediate repercussion on the buying public.

Music lovers, for instance have been forced to return CDs of their favourite artists because the media would not play on their computer systems. Take the case of the soundtrack for the movie *The Fast and the Furious*. Media giant, Universal, tapped a copy-protection technology to lock up on ordinary CD audio files. These songs, which appeared to be MP3s or related to MP3s, could not be played through common media players such as Windows Media Player, Winamp or

have opted to use copy protection schemes such as Macrovision's SafeDisc and many a gamer has found himself on the receiving end of the problems caused by such solutions: games bought off store shelves refuse to run on their computer systems because they have older CD-ROM drives. The authentication process also takes up system resources and slows down a gaming session. Some of the games have also been known to cause frequent system crashes.

Situations like these are a common occurrence and will continue to plague the buying public. It is thus the end-user who finds himself caught between the horses of pirates and copy-protection schemes. But perhaps the key lies with the consumer himself—if he is made aware of the evils of piracy and is encouraged by the content owners to buy the real deal, there will be no need to employ schemes with potential pitfalls. Until that happens, the industries involved will always try to find solutions to protect their content and keep the customers happy...and they will always fail! 

AHMED SHAIKH

info The Interactive Digital Software Association estimated that the piracy of games amounted to a loss of over \$3.2 billion worldwide